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Ex. No: 2Statistical Measures – Entropy, Cross Entropy and Perplexity

Implement Statistical measures such as Entropy, Cross Entropy and Perplexity to analyze the text for the given Datasets

Datasets:
- 1. Brown Corpus
- 2. Reuters Corpus
- 3. Newsgroups Corpus
- 4. Universal Dependencies English Tree Bank

Aim
To implement and analyze statistical measures — Entropy, Cross Entropy, and Perplexity — on standard NLP corpora to quantitatively evaluate the predictability and complexity of natural language text.

Procedure/Pseudocode
- 1. Load each corpus (Brown, Reuters, Newsgroups, UD English Tree Bank) and preprocess the text.
- 2. Build a unigram probability distribution from word frequencies using NLTK's FreqDist.
- 3. Entropy Calculation: Compute Shannon entropy $H(X) = -\Sigma P(x) \log_2 P(x)$ over all unique words.
- 4. Cross Entropy Calculation: Use a reference distribution (Brown) and compute cross entropy $H(P, Q) = -\Sigma P(x) \log_2 Q(x)$ for each target corpus.
- 5. Perplexity Calculation: Compute perplexity as PP(W) = 2^H where H is the cross entropy per word.
- 6. Compare entropy values across corpora to assess vocabulary diversity.
- 7. Visualize entropy, cross entropy, and perplexity values using bar charts.

Libraries Used
- 1. nltk – for corpus loading and frequency distribution
- 2. numpy – for logarithmic and mathematical computations
- 3. math – for entropy and perplexity computation
- 4. matplotlib – for visualization of statistical measures
- 5. collections.Counter – for token frequency counting

Test Cases

- 1. Brown Corpus: Vocabulary size ≈ 56,000 tokens → Entropy ≈ 9.8 bits → Perplexity ≈ 890
- 2. Reuters Corpus: Vocabulary size ≈ 41,000 tokens → Entropy ≈ 9.4 bits → Perplexity ≈ 675
- 3. Cross entropy between Brown (reference) and Newsgroups: Higher value indicates domain divergence
- 4. Uniform distribution test: For vocabulary of size V, max entropy = $\log_2(V)$ bits

Results

Entropy values varied across corpora, with more diverse general corpora (Brown) showing higher entropy than domain-specific ones (Reuters). Cross entropy was lowest when the reference and target corpora were from similar domains, confirming linguistic similarity. Perplexity scores were consistent with intuitions about text predictability.